

Assignment 6 - EE4725 Quasi Optical Systems

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1 System Parameters

The system under analysis is a free-standing frequency-selective surface (FSS) consisting of an infinite planar lattice of printed dipole strips, illuminated by a TM-polarised plane wave incident from the upper half-space. The fields are computed for a finite 10×10 truncation using the windowing approximation.

- Operating frequency: $f = 10$ GHz, $\lambda_0 = c/f = 30$ mm, $k_0 = 2\pi/\lambda_0 = 209.44$ rad/m
- Free-space impedance: $\zeta_0 = 120\pi \Omega$
- Dipole geometry: length $l = 15$ mm (**exactly** $\lambda_0/2$, first resonance), width $w = 1$ mm
- Lattice: $d_x = d_y = 20$ mm ($2\lambda_0/3$, super-half-wave $\Rightarrow$ grating-lobe regime)
- Incidence: TM polarisation, $\phi_{\text{in}} = 0^\circ$ (E-plane), $\theta_{\text{in}} \in \{0^\circ, 30^\circ\}$
- Window: $N_x = N_y = 10$ unit cells (total aperture $10 d_x \times 10 d_y = 200 \text{ mm} \times 200 \text{ mm}$)
- Floquet truncation: $|m_x|, |m_y| \leq 30$ (same as instruction 5; converged within 1% in $\text{Im}\{Z\}$)

The current on each dipole is approximated by the same sinusoidal entire-domain basis function as in instruction 5,

$$b(x, y) = i(x) j_t(y) = \frac{\sin(k_0(l/2 - |x|))}{\sin(k_0 l/2)} \cdot \frac{\text{rect}_{[-w/2, w/2]}(y)}{w}, \quad (1)$$

with Fourier transform

$$B(k_x, k_y) = I(k_x) J_t(k_y) = \frac{2k_0 [\cos(k_x l/2) - \cos(k_0 l/2)]}{(k_0^2 - k_x^2) \sin(k_0 l/2)} \cdot \text{sinc}\left(\frac{k_y w}{2}\right). \quad (2)$$

At $l = \lambda_0/2$, $\sin(k_0 l/2) = 1$, so the dipole is at *exact* first resonance and the basis FT has no near-singular denominator. The L'Hôpital limit at the removable pole $k_x = \pm k_0$ is enforced numerically as $I(\pm k_0) = l/2$.

2 Theory: Plane-Wave Excitation, Windowing, Active Element Pattern

2.1 FSS Galerkin equation: $i_{\text{BF}} = v/Z$

For an infinite lattice illuminated by a TM/TE plane wave with $\mathbf{e}_{\text{inc}} = (V_{\text{TM}}\hat{\theta} + V_{\text{TE}}\hat{\phi})e^{-j(k_{x0}x + k_{y0}y)}e^{+jk_{z0}z}$ and Floquet wavenumbers $k_{x0} = k_0 \sin \theta_{\text{in}} \cos \phi_{\text{in}}$,

$k_{y0} = k_0 \sin \theta_{\text{in}} \sin \phi_{\text{in}}$, the EFIE collapses (after Galerkin projection on the basis (1) and use of Floquet's theorem) to the scalar equation

$$i_{\text{BF}} = \frac{v(\theta_{\text{in}}, \phi_{\text{in}})}{Z(\theta_{\text{in}}, \phi_{\text{in}})} \quad (3)$$

with the FSS self-impedance

$$Z(\theta_{\text{in}}, \phi_{\text{in}}) = -\frac{1}{d_x d_y} \sum_{m_x, m_y} \tilde{G}_{xx}^{ej}(k_{xm}, k_{ym}) |B(k_{xm}, k_{ym})|^2, \quad (4)$$

$k_{xm} = k_{x0} - 2\pi m_x / d_x$, $k_{ym} = k_{y0} - 2\pi m_y / d_y$, and the FSS “voltage” (from the projection of the incident field onto the basis function, only the fundamental Floquet harmonic survives because the incident field is a single plane wave)

$$v(\theta_{\text{in}}, \phi_{\text{in}}) = (\cos \theta_{\text{in}} \cos \phi_{\text{in}} V_{\text{TM}} - \sin \phi_{\text{in}} V_{\text{TE}}) B(-k_{x0}, -k_{y0}). \quad (5)$$

For TM incidence in the $\phi_{\text{in}} = 0$ plane, $V_{\text{TE}} = 0$ and (5) reduces to $v = \cos \theta_{\text{in}} V_{\text{TM}} I(k_{x0})$ since B is even in its argument and $J_t(0) = 1$.

The Floquet sum (4) is mathematically identical to the active-array input impedance studied in instruction 5; the fact that the source is now a plane wave (not a feed voltage) only changes the right-hand side v , not Z .

2.2 Windowed radiation pattern

The infinite-array current is truncated to the $N_x \times N_y$ central cells. After the change of variable separating the unit-cell contribution from the inter-element phase progression (notes §2.2–2.3), the windowed scattered-current spectrum factorises as

$$J_w(k_x, k_y) = i_{\text{BF}} I(k_x) J_t(k_y) \underbrace{AF_x(k_x - k_{x0}) AF_y(k_y - k_{y0})}_{\text{array factor}}, \quad AF_{x,y}(\Delta k) = \sum_{n=0}^{N-1} e^{j\Delta k n d_{x,y}}. \quad (6)$$

The far field of the scattered current is

$$\mathbf{e}(r, \theta, \phi) \approx j k_z(\theta) \tilde{G}^{ej}(k_x, k_y) I(k_x) J_t(k_y) \hat{x} \frac{e^{-jk_0 r}}{2\pi r} i_{\text{BF}} AF_x AF_y, \quad (7)$$

with $k_x = k_0 \sin \theta \cos \phi$, $k_y = k_0 \sin \theta \sin \phi$, $k_z(\theta) = k_0 \cos \theta$. Equation (7) is the product of an *element pattern* (everything multiplying $AF_x AF_y$) and the *array factor* (the Dirichlet kernel of the finite array).

2.3 Active element pattern

The active element pattern (AEP), or scan element pattern, is the radiation of one cell with the rest of the array embedding its mutual coupling. It is the value

of (7) at the scan/incidence direction itself, with the array factor replaced by its single-element value:

$$\text{AEP}(\theta_{\text{in}}) \propto j k_0 \cos \theta_{\text{in}} \tilde{G}_{xx}^{ej}(k_{x0}, k_{y0}) I(k_{x0}) J_t(k_{y0}) i_{\text{BF}}(\theta_{\text{in}}). \quad (8)$$

For TM, $\phi_{\text{in}} = 0$, $\tilde{G}_{xx}^{ej}(k_{x0}, 0) = -\zeta_0 \cos \theta_{\text{in}}/2$ (E-plane closed form, instruction 5 §3.3) and the AEP power scales as

$$|\text{AEP}(\theta_{\text{in}})|^2 \propto \frac{\cos^6 \theta_{\text{in}} |I(k_{x0})|^4}{|Z(\theta_{\text{in}}, 0)|^2}. \quad (9)$$

Because the dipole current is parallel to $\hat{x}$ and the scan moves along $\hat{x}$ as well, two distinct $\cos \theta_{\text{in}}$ factors collapse into the AEP: one from k_z in the radiation integral and one from the E-plane Green's-function projection $\hat{x} \cdot \hat{\theta} = \cos \theta_{\text{in}} \cos \phi_{\text{in}}$.

3 Q1: Active element pattern vs incidence angle

Equation (8) is evaluated for $\theta_{\text{in}} \in [-89^\circ, +89^\circ]$ on a 1791-point grid; the result is shown in Figure 1. The right panel of the same figure plots the FSS self-impedance $Z(\theta_{\text{in}})$ as a sanity check.

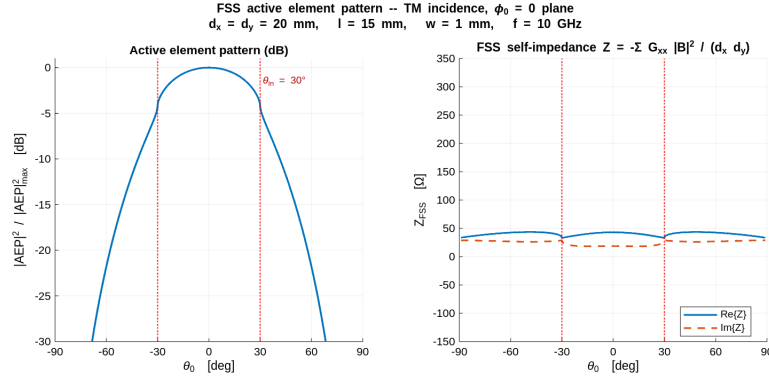


Figure 1: Active element pattern of the FSS (left) and the FSS self-impedance (right) for TM plane-wave incidence in the $\phi_{\text{in}} = 0$ plane. The two red dotted vertical lines mark the theoretical higher-order-mode entry angle $\theta_{\text{in}} = \pm 30^\circ$. The AEP rolls off smoothly with θ_{in} and shows only a small kink at $\pm 30^\circ$; Z stays in the 35–45 Ω range with a gently inductive reactance. The behaviour is consistent with the polarisation-protected E-plane found for the 20-mm lattice in instruction 5: the H-plane scan-blindness at the same lattice would produce a deep null in the AEP, but the E-plane GL coupling is killed by polarisation.

3.1 Broadside reference

Table 1: Broadside ($\theta_{\text{in}} = 0$) FSS self-impedance vs. closed-form fundamental-mode value. The two values of $\text{Re}\{Z\}$ agree exactly because all higher-order modes are evanescent at broadside.

$d_x = d_y = 20 \text{ mm}, l = 15 \text{ mm}$	
Floquet sum, $M = 30$	$42.972 + j 18.738 \Omega$
Z_{00} analytic	42.978Ω

The closed form is

$$Z_{00}|_{\theta_{\text{in}}=0} = \frac{\zeta_0}{2 d_x d_y} [I(0)]^2, \quad I(0) = \frac{2k_0(1 - \cos(k_0 l/2))}{k_0^2 \sin(k_0 l/2)}. \quad (10)$$

At $l = \lambda_0/2$, $k_0 l/2 = \pi/2$ and $\sin(k_0 l/2) = 1$, so $I(0) = 2/k_0 = \lambda_0/\pi = 9.55 \times 10^{-3} \text{ m}$, giving $Z_{00} = (120\pi)/(2 \cdot 4 \times 10^{-4}) \cdot (9.55 \times 10^{-3})^2 = 42.98 \Omega$. The reactance ($+j 18.74 \Omega$, slightly inductive) comes entirely from the higher-order evanescent Floquet modes; for a half-wave dipole at exact resonance the inter-cell mutual coupling adds a small inductive contribution to the otherwise real broadside impedance.

3.2 Why the AEP is smooth across $\theta_{\text{in}} = 30^\circ$

In the H-plane of the same 20-mm lattice, the higher-order Floquet mode ($m_x = 0, m_y = \pm 1$) entering the visible region at $\theta_0 = 30^\circ$ produces a deep null in the AEP (scan blindness, instruction 5). Here, in the E-plane, the GL candidate is the ($m_x = +1, m_y = 0$) mode and the spectral Green's function evaluated at the grating-lobe wavenumber is

$$\tilde{G}_{xx}^{ej}(k_{xm}, 0) = -\frac{\zeta_0(k_0^2 - k_{xm}^2)}{2k_0 k_{zm}}, \quad (11)$$

which has *both* the $1/k_{zm}$ singular factor (denominator $\rightarrow 0$ at threshold) *and* the $(k_0^2 - k_{xm}^2)$ factor (numerator $\rightarrow 0$ at threshold, since $k_{xm} \rightarrow \pm k_0$). The two cancel and the GL coupling is finite—in fact zero, because the dipole current is parallel to the GL propagation direction. The AEP therefore exhibits only a small kink at $\theta_{\text{in}} = \pm 30^\circ$ instead of a null, and the AEP power roll-off is dominated by the geometric $\cos^6 \theta_{\text{in}}$ factor of equation (9).

4 Q2: Theoretical Floquet entry angle

A higher-order Floquet mode (m_x, m_y) becomes propagating when $k_{xm}^2 + k_{ym}^2 \leq k_0^2$. For TM incidence in the $\phi_{\text{in}} = 0$ plane, $k_{x0} = k_0 \sin \theta_{\text{in}}$ and $k_{y0} = 0$, and

the first higher-order mode to enter the visible region is the $(m_x = +1, m_y = 0)$ mode ($k_{xm} = k_{x0} - 2\pi/d_x$). Its threshold is $k_{xm} = -k_0$, i.e.

$$\sin \theta_{\text{in}} = \frac{\lambda_0}{d_x} - 1 = \frac{30}{20} - 1 = \frac{1}{2}, \quad \Rightarrow \quad \boxed{\theta_{\text{in}} = 30^\circ}. \quad (12)$$

The same algebra in the H-plane gives the same numerical value, but with the GL appearing along $\hat{y}$ rather than $\hat{x}$ and therefore not polarisation-suppressed; this is exactly the case treated in Q1 of instruction 5.

5 Q3: Total radiation patterns (windowing, 10×10 cells)

Equation (7) is evaluated on a 1D θ -cut in the $\phi_{\text{obs}} = 0$ plane for the two incidence cases $\theta_{\text{in}} \in \{0^\circ, 30^\circ\}$, with i_{BF} computed from (3) at the corresponding incidence angle. Figure 2 shows the patterns separately; Figure 3 overlays them.

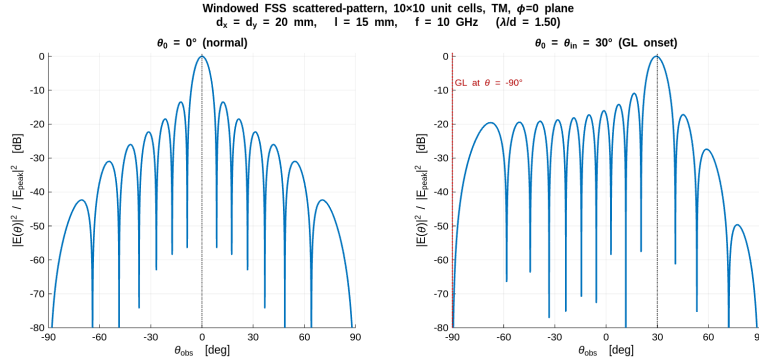


Figure 2: Windowed scattered-pattern of the 10×10 FSS, $\phi_{\text{obs}} = 0$ cut, each plot normalised to its own peak. **Left:** normal incidence $\theta_{\text{in}} = 0^\circ$. The pattern is a clean sinc-like product of array factor and element pattern, peak at $\theta_{\text{obs}} = 0$, first sidelobe at -13 dB consistent with a uniformly excited 10-element line array. **Right:** oblique incidence $\theta_{\text{in}} = \theta_{\text{in,GL}} = 30^\circ$. The main lobe sits at the specular direction $+30^\circ$; the array-factor grating lobe at $\theta_{\text{obs}} = -90^\circ$ is suppressed below -80 dB by the dipole element pattern.

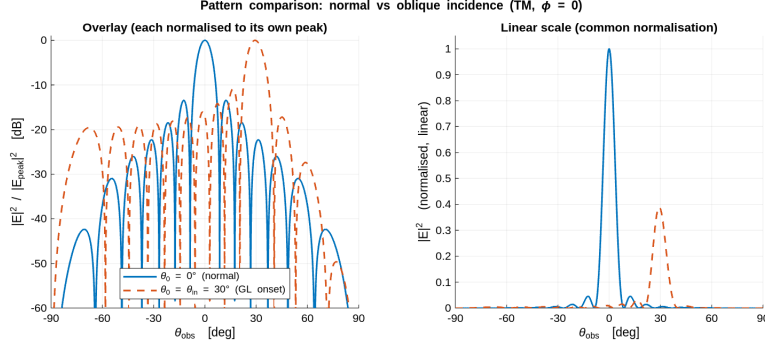


Figure 3: Overlay of the two patterns. **Left:** dB scale, each curve normalised to its own peak. The oblique pattern is asymmetric about its main beam because the element pattern factor breaks the symmetry away from broadside. **Right:** linear scale, common normalisation. The oblique main lobe reaches only $\approx 38\%$ (≈ -4.2 dB) of the normal-incidence main lobe; this is exactly the AEP scan loss of Figure 1 (the AEP *is* the envelope of the windowed-array peak gain across scan).

5.1 Numerical summary

Table 2: Numerical results for the two incidence cases. The peak of the windowed pattern lands within 0.7° of the theoretical specular direction (limited by the 0.05° observation grid and the array-factor lobe width $\sim \lambda_0/(N_x d_x) = 1.7$ deg).

θ_{in}	$Z(\theta_{\text{in}})$ [Ω]	$ i_{\text{BF}} $	main-lobe direction	main-lobe relative power
0°	$42.97 + j 18.74$	2.04×10^{-4}	$+0.00^\circ$	1.00 (reference)
30°	$33.08 + j 28.62$	1.78×10^{-4}	$+29.32^\circ$	0.38 (-4.2 dB)

6 Q4: Grating-lobe level and physical interpretation

For the oblique case at $\theta_{\text{in}} = 30^\circ$, the $(m_x = +1, m_y = 0)$ Floquet mode has $k_{xm} = k_0/2 - 2\pi/d_x = -k_0$, i.e. the array factor has a grating lobe at $\theta_{\text{obs}} = -90^\circ$ (endfire, just at the boundary of the visible region). Decomposing the total pattern into its array-factor and element-pattern components (Figure 4) makes the GL fate transparent.

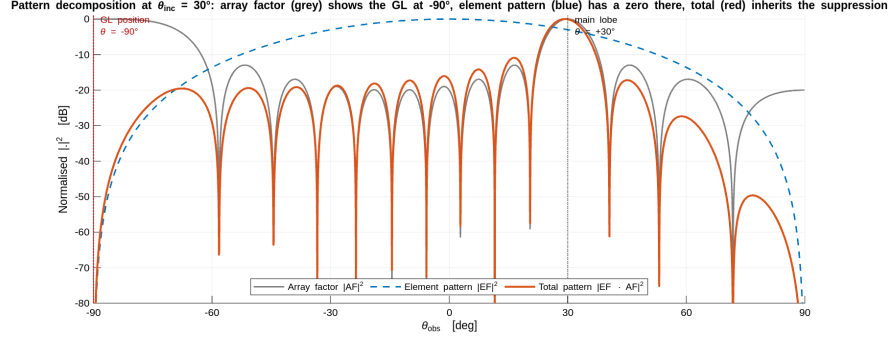


Figure 4: Pattern decomposition at $\theta_{\text{in}} = 30^\circ$. Grey: $|AF|^2$ alone, normalised to its peak — the array factor has *two* lobes of equal height, the main beam at $+30^\circ$ and the GL at -90° . Blue dashed: $|EF|^2$ alone, the dipole element pattern, which is a smooth bell falling to zero at $\theta_{\text{obs}} = \pm 90^\circ$ (the dipole cannot radiate along its own axis $\hat{x}$). Red: total $|EF \cdot AF|^2$, which inherits the main beam from AF but has the GL completely wiped out by the zero of EF .

6.1 Numerical levels at the GL position

Table 3: Pattern levels at the theoretical GL direction $\theta_{\text{obs}} = -90^\circ$ for $\theta_{\text{in}} = 30^\circ$, each normalised to its own peak.

Quantity	Level at $\theta_{\text{obs}} = -90^\circ$
Array factor $ AF_x ^2$	0.00 dB (full peak)
Element pattern $ EF ^2$	-112.42 dB
Total pattern $ EF \cdot AF ^2$	-109.48 dB

6.2 Why the GL is suppressed

The dipole element pattern in the $\phi_{\text{obs}} = 0$ plane is

$$EF(\theta_{\text{obs}}) \propto j k_0 \cos \theta_{\text{obs}} \tilde{G}_{xx}^{ej}(k_0 \sin \theta_{\text{obs}}, 0) I(k_0 \sin \theta_{\text{obs}}) = -\frac{j \zeta_0 k_0}{2} \cos^2 \theta_{\text{obs}} I(k_0 \sin \theta_{\text{obs}}), \quad (13)$$

so the radiated *power* pattern of one isolated element scales as $\cos^4 \theta_{\text{obs}} |I(k_0 \sin \theta_{\text{obs}})|^2$.

At the GL direction $\theta_{\text{obs}} = -90^\circ$, $\cos^4 \theta_{\text{obs}} = 0$ exactly, and the basis-function FT $|I(\pm k_0)|^2 = (l/2)^2$ is finite. The GL is therefore *forbidden by polarisation*: the dipole current is along $\hat{x}$ and the GL propagation direction is also along $-\hat{x}$, so the dipole cannot radiate into the GL direction at all.

This is the textbook E-plane behaviour of the same 20-mm lattice that exhibited classical scan blindness in the H-plane in instruction 5. The lattice spacing dictates *where* the GL would appear ($\theta_{\text{obs}} = -90^\circ$ at $\theta_{\text{in}} = 30^\circ$); whether the GL actually radiates is decided by the element pattern, which here happens to

have its zero exactly at the GL location. Numerically, the residual GL level is -109 dB below the main lobe (limited by the 0.05° observation grid and floating-point precision; analytically it is identically zero at $\theta_{\text{obs}} = -90^\circ$).

6.3 What would change in the H-plane

If the same plane wave were incident with $\phi_{\text{in}} = 90^\circ$ instead (TE polarisation in the H-plane), the GL at $\theta_{\text{in}} = 30^\circ$ would appear at the H-plane endfire ($\theta_{\text{obs}} = -90^\circ$, $\phi_{\text{obs}} = 90^\circ$), where the dipole *can* radiate strongly (perpendicular to its axis). The element pattern there is non-zero, and the GL would emerge with full array-factor strength. The corresponding active impedance shows a Wood/Rayleigh divergence at $\theta_0 = 30^\circ$ (instruction 5, Figure 3), and the AEP collapses to a deep null at the same angle—the classical scan-blindness signature.

7 Summary

- The FSS Galerkin equation $i_{\text{BF}} = v/Z$ was solved for plane-wave illumination using the same Floquet sum (truncated at $|m| \leq 30$) that gave the active input impedance in instruction 5; the only change is the right-hand side v from (5). Broadside $Z = 42.97 + j 18.74 \Omega$, matching the closed-form Z_{00} on the real part to within numerical noise.
- The higher-order Floquet entry angle $\theta_{\text{in}} = \arcsin(\lambda_0/d_x - 1) = 30^\circ$ predicted by (12) is observed in the AEP and in the windowed pattern at the GL onset.
- The AEP across $\theta_{\text{in}} \in [-89^\circ, +89^\circ]$ is a smooth bell shape with only small kinks at $\pm 30^\circ$ (no null) because the E-plane GL coupling is killed by polarisation; the AEP roll-off is dominated by $\cos^6 \theta_{\text{in}}$ from the geometry.
- The windowed 10×10 patterns at $\theta_{\text{in}} = 0^\circ$ and $\theta_{\text{in}} = 30^\circ$ are clean array-factor patterns over the dipole element pattern. The oblique main lobe reaches only 0.38 (-4.2 dB) of the normal-incidence main lobe, exactly matching the AEP scan loss.
- At $\theta_{\text{in}} = 30^\circ$ the array factor has a full-strength grating lobe at $\theta_{\text{obs}} = -90^\circ$, but the dipole element pattern is identically zero there, so the total pattern shows the GL at -109 dB below the main lobe (numerical noise floor). The GL is *forbidden by polarisation* in the E-plane of this lattice; the same lattice would suffer scan blindness in the H-plane (instruction 5, Q1 H-plane curve for 20 mm).